

LAB MANUAL



COM682 – Lab 01

Azure Account Creation

**Faculty of Computing, Engineering
and The Built Environment**

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1. What is Cloud Computing?

Let us imagine you have a brilliant idea about a new app. For simplicity, let us assume this app is a new food delivery app. The app should have the same experience no matter what device the user uses, mobile or computer, and it will need to give the same experience to the user across all devices. The app will need a data server that stores user information such as login details, orders, favourite restaurants, to name a but few. The data should be secure, and most importantly, data reliable. The server should be up 24*7. No lack or delay; otherwise, it will cost you an unhappy user or abandonment of the app. Generally, the data server is a lot harder to maintain and very expensive.

Furthermore, with the growth of the user number we need more space and more power and regularly need to upgrade the data server. Moreover, if the app failed after release and was not as successful as anticipated, all the expensive is nearly impossible to restore. To solve these issues, we can use Cloud computing.

Cloud computing is the on-demand availability of computer resources, including the data server, machine learning, user authentication, reliability, security and much more. Furthermore, you only need to pay for the services you used. So, if your app is not much success, you can stop using the services at once and save yourself a fortune. The other most significant advantage for Cloud computing is that you are using resources virtually, and you are not bothered with the physical maintenance of the server. Due to Cloud computing in today's world, apps and software's are far easier to maintain, and users have the same experience on different platforms. The developers release the updates more often, which was impossible before Cloud computing. In this course, we will use Microsoft Azure. Azure is a cloud computing platform provided by Microsoft.

2. Azure

Microsoft Azure, or simply Azure, is a cloud-based computing platform developed by Microsoft. Azure includes services like building, testing, deploying and managing cloud base applications. Other platforms which provided similar services include Amazon Web Services (AWS) and Google Cloud Platform.

However, in this module, we will use Azure. Azure provides \$100 free credit to students for 12 months. It can be used along with the free services provided by Azure for students. The following guide will provide step by step instructions for creating an Azure Student Account.

3. Identity and Access Management (IAM)

The first thing that comes to mind about Cloud computing is security. Security for the resources, data and malicious attacks on the sensitive information. Azure provided identity and access management or short as IAM. IAM is based on four major components.

- (1) Secure access to your resources.
- (2) Protect applications, devices and data.
- (3) Defend against malicious login attempts.
- (4) Safeguard credentials.

4. What is Azure Active Directory?

Azure Active Directory (Azure AD) is Microsoft's cloud-based identity and access management service, which helps your employees sign in and access resources in:

- 1) External resources, such as Microsoft 365, the Azure portal, and thousands of other SaaS applications.
- 2) Internal resources, such as apps on your corporate network and intranet, along with any cloud apps developed by your own organization. For more information about creating a tenant for your organization, see QuickStart: Create a new tenant in Azure Active Directory.

5. Who uses Azure AD?

Azure AD is intended for:

- 1) **IT admins.** As an IT admin, you can use Azure AD to control your apps and app resources based on your business requirements. For example, you can use Azure AD to require multi-factor authentication when accessing critical organizational resources. Additionally, you can use Azure AD to automate user provisioning between your existing Windows Server AD and your cloud apps, including Microsoft 365. Finally, Azure AD gives you powerful tools to automatically help protect user identities and credentials and to meet your access governance requirements. To get started, sign up for a free 30-day Azure Active Directory Premium trial.
- 2) **App developers.** As an app developer, you can use Azure AD as a standards-based approach for adding a single sign-on (SSO) to your app, allowing it to work with a user's pre-existing credentials. Azure AD also provides APIs that can help you build personalized app experiences using existing organizational

data. To get started, sign up for a free 30-day Azure Active Directory Premium trial. For more information, you can also see Azure Active Directory for developers.

3) **Microsoft 365, Office 365, Azure, or Dynamics CRM Online subscribers.** As a subscriber, you're already using Azure AD. Each Microsoft 365, Office 365, Azure, and Dynamics CRM Online tenant is automatically an Azure AD tenant. You can immediately start to manage access to your integrated cloud apps.

6. Cloud vs On-Premises Software

Cloud vs on-premises essential difference is where it resides. On-premises software is installed on the local computer or organization servers. At the same time, cloud software is hosted on the vendor's (i.e. Azure, AWS) servers and accessed via web browsers.

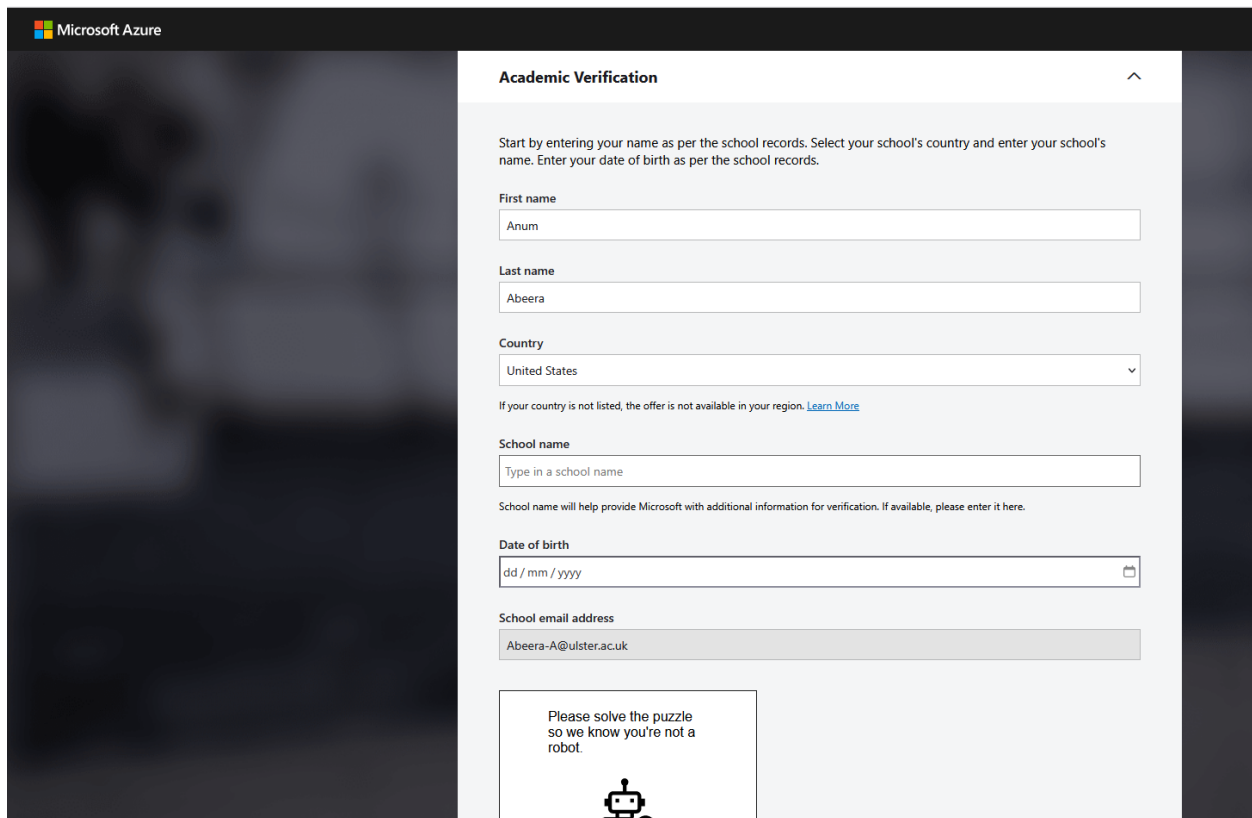
7. Creating Azure Student Account

Step 1:

Please click on the following link.

<https://azureforeducation.microsoft.com/devtools>

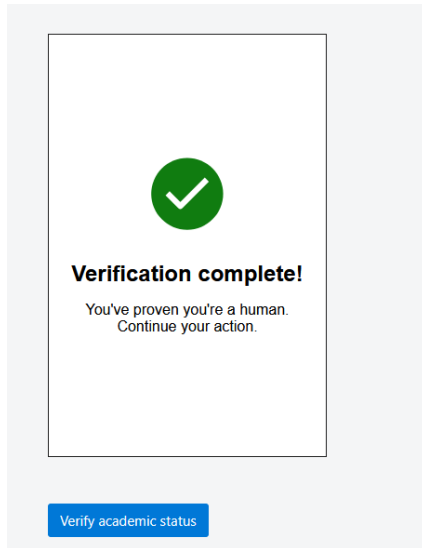
- It should ask you for Multi Factor Authentication.
- After authenticating through your Microsoft Authenticator app, you will see following screen:



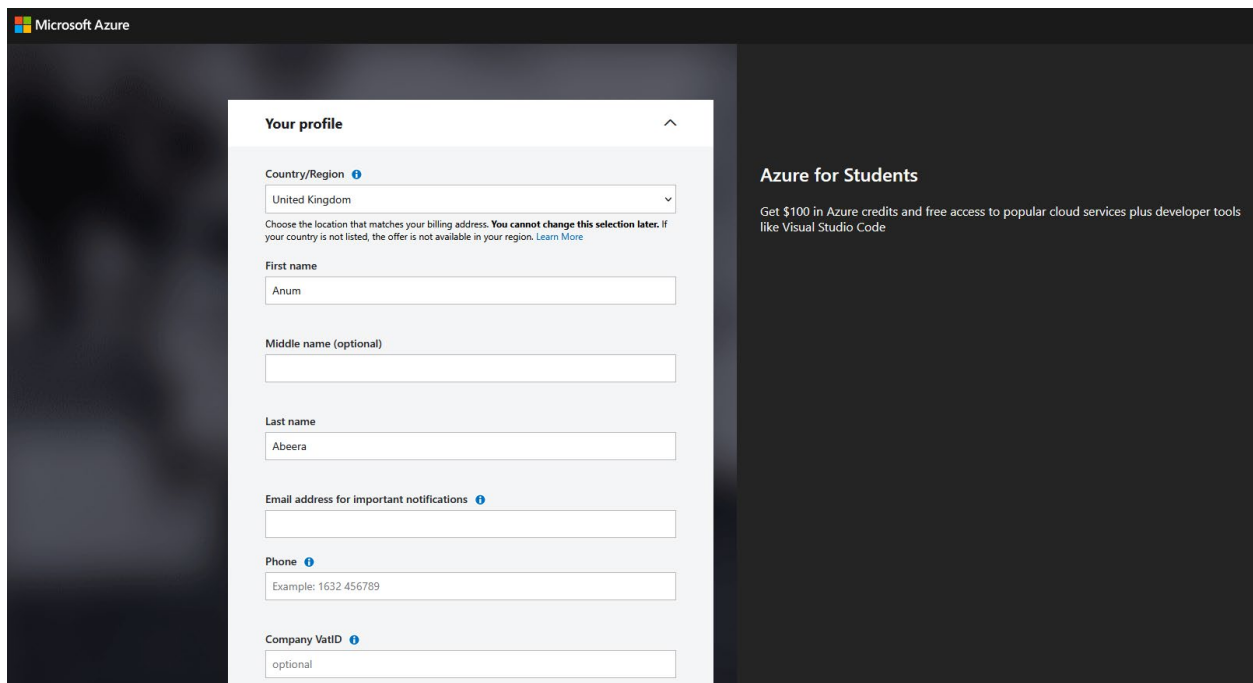
The screenshot shows the 'Academic Verification' page on the Microsoft Azure portal. The page has a dark header with the 'Microsoft Azure' logo. The main content area is light gray and contains the following fields and instructions:

- Academic Verification** (Title)
- Instructions: "Start by entering your name as per the school records. Select your school's country and enter your school's name. Enter your date of birth as per the school records."
- First name**: Text input field containing "Anum".
- Last name**: Text input field containing "Abeera".
- Country**: Dropdown menu showing "United States". Below it, a note says: "If your country is not listed, the offer is not available in your region. [Learn More](#)".
- School name**: Text input field containing "Type in a school name". Below it, a note says: "School name will help provide Microsoft with additional information for verification. If available, please enter it here."
- Date of birth**: Date picker field showing "dd / mm / yyyy".
- School email address**: Text input field containing "Abeera-A@ulster.ac.uk".
- Robot check**: A box with the text "Please solve the puzzle so we know you're not a robot." and a small robot icon.

Fill the form and click on :

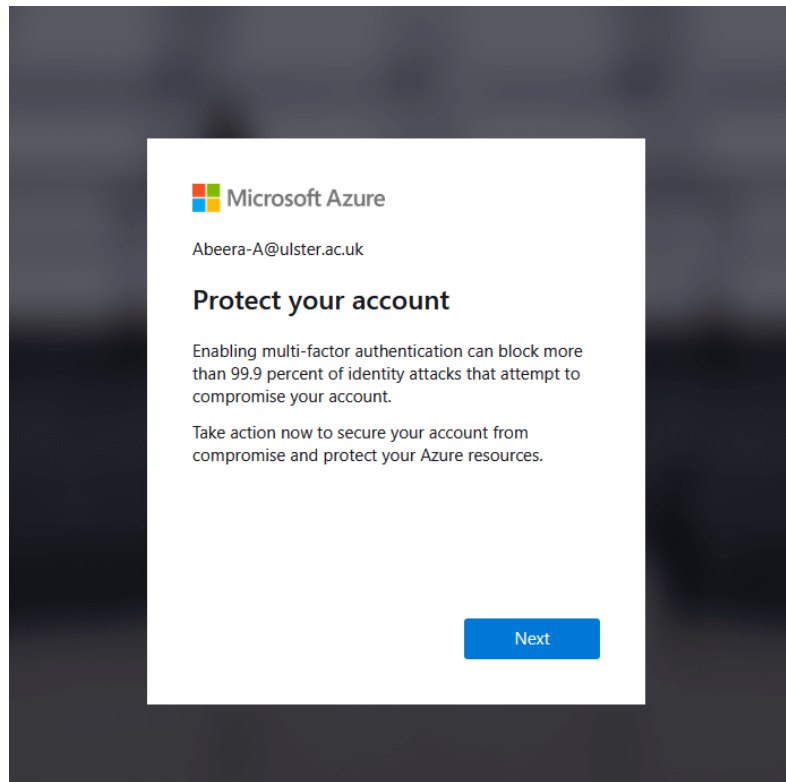


Once your account is confirmed, you will see the screen to get \$100 in Azure Credit.

A screenshot of the Microsoft Azure "Your profile" form. The form is titled "Your profile" and has a back arrow icon. It contains several input fields: "Country/Region" with a dropdown menu showing "United Kingdom"; "First name" with the text "Anum"; "Middle name (optional)" with an empty field; "Last name" with the text "Abeera"; "Email address for important notifications"; "Phone" with an example number "1632 456789"; and "Company VatID" with the text "optional". To the right of the form, there is a section titled "Azure for Students" which states: "Get \$100 in Azure credits and free access to popular cloud services plus developer tools like Visual Studio Code".

- Enter your Ulster Email address
- Enter your address, Post Code
- Make sure to check the 'Microsoft Customer Agreement' , and click Signup.

You will see the following screen to verify the Multi Factor Authentication process:



If everything is successful then you will see the following screen:

Home > Education | Overview

I want to build an OpenAI application with JS | What resources are available for students? | Generate an outline to ramp up on Azure

Get started Overview

Overview

- > Learning resources
- > Need help?

Student offer details

Available credits
\$100 out of \$100

Days until credit expires
365
Expires on 20/09/2026

September costs
\$0.00

[View cost details](#)

Popular solutions

[Deploy a Docker container](#)
Create simple containers to host apps,

[Create your first Node.js app](#)
Build and deploy web, mobile and API-based

[Create and train a Machine Learning model](#)
Train, deploy, automate, manage, and track

[Build and deploy your first website](#)
Automatically publish to web as your code

[Explore all](#)

Free Services

[Azure Virtual Machines – Windows](#)
Use 750 hours of access to B1s virtual

[Azure Blob Storage](#)
Get 5 GB of locally redundant storage (LRS)

[Computer Vision](#)
Receive 5000 AI transactions to process visual

[Azure App Service](#)
Quickly create up to 10 powerful apps with 1

[Explore all](#)

Free software

[SQL Server 2019 Developer](#)

[Machine Learning Server 9.4.7 for Windows](#)

[Visual Studio Enterprise Edition 2022](#)

[Microsoft R Client 9.4.7](#)

[Agents for Visual Studio 2019 \(version 16.0\) Test Agent](#)

[Explore all](#)

Free learning paths

[Data Scientist](#)
Nineteen learning paths with 75+ hours of content.

[AI Engineer](#)
Nine learning paths with 25+ hours of content.

[Developer](#)
Forty-six learning paths with 110+ hours of content.

[DevOps Engineer](#)
Eight learning paths with 50+ hours of content.

[Explore all](#)

Resources

[Get started guide for Azure developers](#)

[Pricing calculator](#)

[Optimize your cloud investment with cost management](#)

[Explore student hub](#)

[FAQs](#)

Congratulations you have for free \$100 Azure credit to be used.

For future use, simply use this link to access Microsoft Azure account:

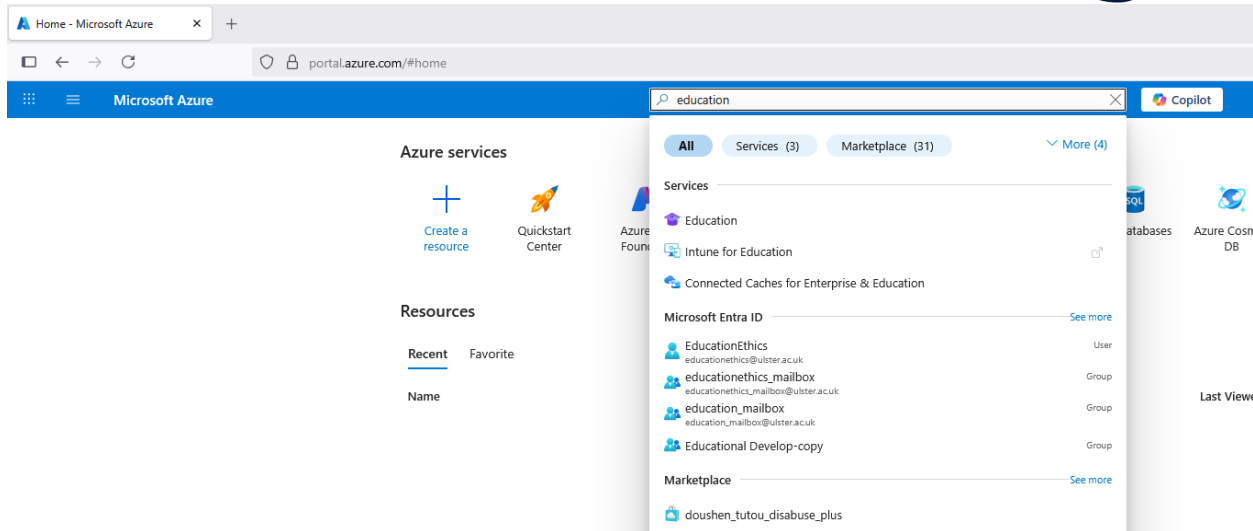
<https://portal.azure.com/#home>

Step 2:

Please sign out and login to Azure again using the link: <https://portal.azure.com/#home>

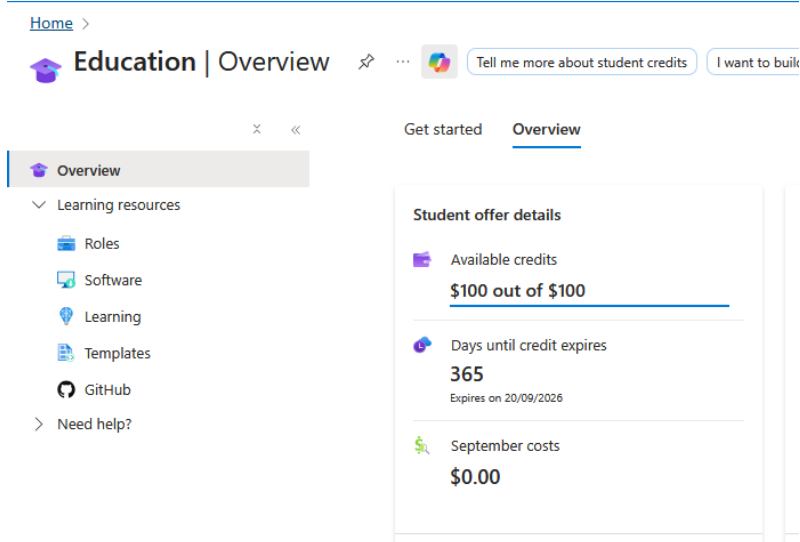
Please click on Sign In. And login with your **ulster** email ID.

In search bar type education, to search for your education page:



Step 3:

Click on 'Learning Resources' in the left pane. And then click Software



(https://portal.azure.com/#blade/Microsoft_Azure_Billing/FreeServicesBlade).

Microsoft Azure

Search resources, services, and docs (G+J)

Home >

Free services

Services free for 12 months with the Azure free account

Services that are included for free with your Azure free account. You can use these services within the limits listed below without getting charged. To learn more, see the [Azure free account FAQ](#).

Windows Virtual Machine COMPUTE 750 hours B1S Create Windows virtual machines with on-demand capacity in seconds. Learn more Create	Linux Virtual Machine COMPUTE 750 hours B1S Create Linux virtual machines with on-demand capacity in seconds. Learn more Create	Azure Managed Disks STORAGE 64 GB x 2 Get premium, secured disk storage for Azure Virtual Machines with simplified management. Learn more Create	Azure Blob Storage STORAGE 5 GB LRS hot block Use massively-scalable object storage for any type of unstructured data. Learn more Create	Azure File Storage STORAGE 5 GB LRS File Storage Migrate to simple, distributed, cross-platform file storage without changing code. Learn more Create	SQL Database DATABASES 250 GB Create an Azure SQL Database that delivers intelligence built-in. Learn more Create
Azure Cosmos DB DATABASES 400 RU/s provisioned throughput With 25 GB of storage Get a fast NoSQL database with open APIs for any scale. Learn more Create	Bandwidth (Data Transfer) NETWORKING 15 GB outbound Transfer data inbound and outbound through our robust network of global data centers. Learn more Create	Anomaly Detector AI + MACHINE LEARNING 20,000 transactions per month Detect anomalies in data to quickly identify and troubleshoot issues. Learn more Create	Computer Vision AI + MACHINE LEARNING 5,000 transactions per month Extract rich information from images to categorize and process visual data. Learn more Create	Content Moderator AI + MACHINE LEARNING 5,000 transactions per month Moderate text and images to provide a safer, more positive user experience. Learn more Create	Custom Vision AI + MACHINE LEARNING 2 projects Easily customize computer vision models for your unique use case. Learn more Create
Face AI + MACHINE LEARNING 30,000 transactions per month Detect and identify people and emotions in images. Learn more Create	Form Recognizer AI + MACHINE LEARNING 500 pages Automate the extraction of text, key/value pairs, and tables from your documents. Learn more Create	Ink Recognizer AI + MACHINE LEARNING 2,000 transactions per month Recognize digital ink content, such as handwriting, shapes, and document layout. Learn more Create	Language Understanding AI + MACHINE LEARNING 10,000 text request transactions Build natural language understanding into apps, bots, and IoT devices. Learn more Create	Personalizer AI + MACHINE LEARNING 50,000 transactions per month Deliver rich, personalized experiences for every user. Learn more Create	QnA Maker AI + MACHINE LEARNING 2,000 days Create a conversational question-and-answer bot from your existing content. Learn more Create
Text Analytics AI + MACHINE LEARNING 5,000 transactions per month Extract information such as sentiment, key phrases, named entities, and language from your text. Learn more Create	Translator Text API AI + MACHINE LEARNING 2,000,000 characters included Add real-time, multi-language text translation to your apps, website, and tools. Learn more Create	Service Bus DATABASES 750 hours, 13 million operations Standard tier base unit Get reliable cloud messaging as a service with simple hybrid integration. Learn more Create	Container Registry CONTAINERS 1 Standard tier registry With 100 GB storage and 10 webhooks Store and manage container images across all types of Azure deployments. Learn more Create	Load Balancer NETWORKING 750 hours, 15 GB data processing, 5 rules Standard Load Balancer Deliver high availability and network performance to your applications. Learn more Create	Archive Storage STORAGE 10 GB LRS storage 10 GB LRS/GRS write and retrieval plus 100 reads Store and manage rarely accessed data with locally redundant storage (LRS) or geo-redundant storage (GRS). Learn more Create
VPN Gateway NETWORKING/SECURITY 750 hours VpnGw1 Gateway Type Create					

Home >

Free services

Services that are included for free with your Azure free account. You can use these services within the limits listed below without getting charged. To learn more, see the [Azure free account FAQ](#).

Face AI + MACHINE LEARNING 30,000 transactions per month Detect and identify people and emotions in images. Learn more Create	Form Recognizer AI + MACHINE LEARNING 500 pages Automate the extraction of text, key/value pairs, and tables from your documents. Learn more Create	Ink Recognizer AI + MACHINE LEARNING 2,000 transactions per month Recognize digital ink content, such as handwriting, shapes, and document layout. Learn more Create	Language Understanding AI + MACHINE LEARNING 10,000 text request transactions Build natural language understanding into apps, bots, and IoT devices. Learn more Create	Personalizer AI + MACHINE LEARNING 50,000 transactions per month Deliver rich, personalized experiences for every user. Learn more Create	QnA Maker AI + MACHINE LEARNING 2,000 days Create a conversational question-and-answer bot from your existing content. Learn more Create
Text Analytics AI + MACHINE LEARNING 5,000 transactions per month Extract information such as sentiment, key phrases, named entities, and language from your text. Learn more Create	Translator Text API AI + MACHINE LEARNING 2,000,000 characters included Add real-time, multi-language text translation to your apps, website, and tools. Learn more Create	Service Bus DATABASES 750 hours, 13 million operations Standard tier base unit Get reliable cloud messaging as a service with simple hybrid integration. Learn more Create	Container Registry CONTAINERS 1 Standard tier registry With 100 GB storage and 10 webhooks Store and manage container images across all types of Azure deployments. Learn more Create	Load Balancer NETWORKING 750 hours, 15 GB data processing, 5 rules Standard Load Balancer Deliver high availability and network performance to your applications. Learn more Create	Archive Storage STORAGE 10 GB LRS storage 10 GB LRS/GRS write and retrieval plus 100 reads Store and manage rarely accessed data with locally redundant storage (LRS) or geo-redundant storage (GRS). Learn more Create
VPN Gateway NETWORKING/SECURITY 750 hours VpnGw1 Gateway Type Create					

Please explore the descriptions and “learn more” elements of these services. This will provide you with a brief overview of the capabilities which Azure offers.



These are free services for 12 months. You can use each service for free within the limitations of the free offering.

For example, at the time of writing, you may create use an Azure SQL instance in the S0 configuration (10DTU/s + 250GB storage) for free, for a year.

Ensure you verify what services are free tier and adhere to the limits/spec.

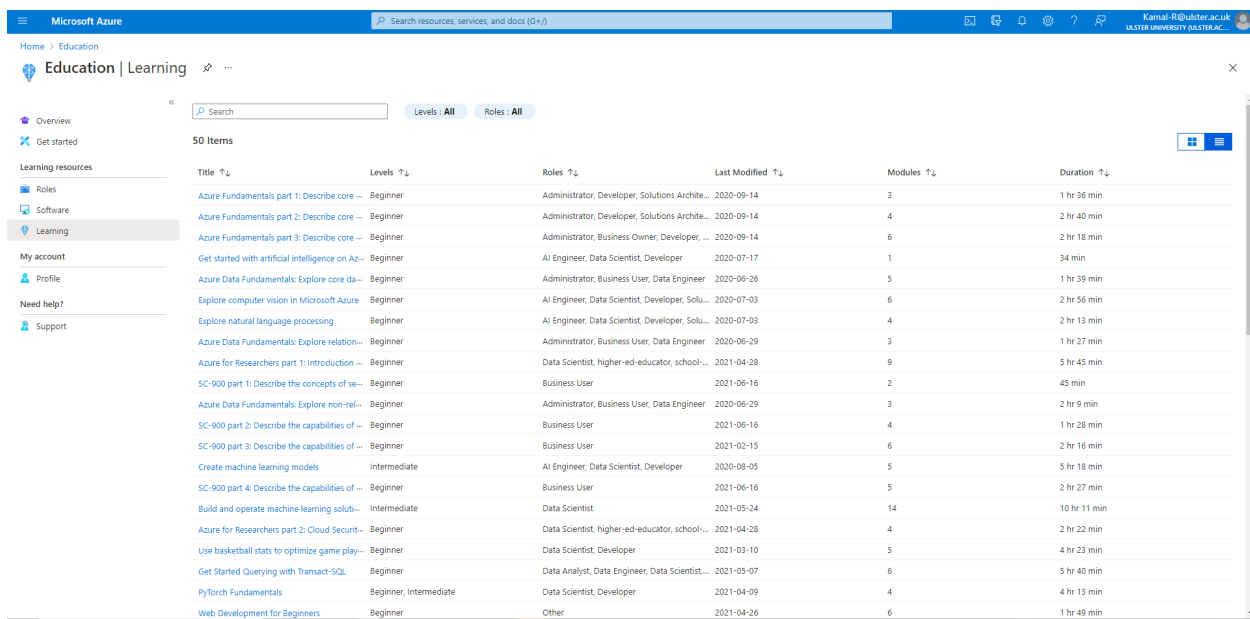
Failure to do so may use your credit, disable your account or incur a bill that you may be liable for.

Please be aware that never leave any service running if you are not using it. Otherwise, your free tier will be consumed, and after that, your free \$100 will be used. If you encounter services which have no free option, then you need to consider how best to leverage your credit. Typically, a serverless instance will be billed at a lower rate.

In free Services, you can see Load balancer, Azure Cosmos, logic apps, functions, Azure Blobs, and SQL database. These services will be used in this module later on. It is strongly recommended to make yourself familiar with these services.

Step 7:

Click back on home and click on education -> learning in the leftmost pane under learning resources.



Title	Levels	Roles	Last Modified	Modules	Duration
Azure Fundamentals part 1: Describe core...	Beginner	Administrator, Developer, Solutions Archite...	2020-09-14	3	1 hr 36 min
Azure Fundamentals part 2: Describe core...	Beginner	Administrator, Developer, Solutions Archite...	2020-09-14	4	2 hr 40 min
Azure Fundamentals part 3: Describe core...	Beginner	Administrator, Business Owner, Developer, ...	2020-09-14	6	2 hr 18 min
Get started with artificial intelligence on Az...	Beginner	AI Engineer, Data Scientist, Developer	2020-07-17	1	34 min
Azure Data Fundamentals: Explore core da...	Beginner	Administrator, Business User, Data Engineer	2020-06-26	5	1 hr 39 min
Explore computer vision in Microsoft Azure	Beginner	AI Engineer, Data Scientist, Developer, Solu...	2020-07-03	6	2 hr 56 min
Explore natural language processing	Beginner	AI Engineer, Data Scientist, Developer, Solu...	2020-07-03	4	2 hr 13 min
Azure Data Fundamentals: Explore relation...	Beginner	Administrator, Business User, Data Engineer	2020-06-29	3	1 hr 27 min
Azure for Researchers part 1: Introduction ...	Beginner	Data Scientist, higher-ed-educator, school-...	2021-04-28	9	5 hr 45 min
SC-900 part 1: Describe the concepts of se...	Beginner	Business User	2021-06-16	2	45 min
Azure Data Fundamentals: Explore non-rel...	Beginner	Administrator, Business User, Data Engineer	2020-06-29	3	2 hr 9 min
SC-900 part 2: Describe the capabilities of ...	Beginner	Business User	2021-06-16	4	1 hr 28 min
SC-900 part 3: Describe the capabilities of ...	Beginner	Business User	2021-02-15	6	2 hr 16 min
Create machine learning models	Intermediate	AI Engineer, Data Scientist, Developer	2020-08-05	5	5 hr 18 min
SC-900 part 4: Describe the capabilities of ...	Beginner	Business User	2021-06-16	5	2 hr 27 min
Build and operate machine learning soluti...	Intermediate	Data Scientist	2021-05-24	14	10 hr 11 min
Azure for Researchers part 2: Cloud Securit...	Beginner	Data Scientist, higher-ed-educator, school-...	2021-04-28	4	2 hr 22 min
Use basketball stats to optimize game play...	Beginner	Data Scientist, Developer	2021-03-10	5	4 hr 23 min
Get Started Querying with Transact-SQL	Beginner	Data Analyst, Data Engineer, Data Scientist...	2021-05-07	6	5 hr 40 min
PyTorch Fundamentals	Beginner, Intermediate	Data Scientist, Developer	2021-04-09	4	4 hr 13 min
Web Development for Beginners	Beginner	Other	2021-04-26	6	1 hr 49 min

Azure provides several free courses from Azure fundamentals to operate and maintain the Azure stack. Please take the Azure fundamental course to know better the basic Azure concepts.

Step 8:

Microsoft provide a range of resources which provide and up to date overview of pricing and budgeting within Azure. These concepts are broadly applicable to most cloud providers.

Please read the following articles, and follow the related activities, in order to understand the pricing process for a variety of different services.

i. Plan and manage costs for Azure App Service (PaaS)

<https://docs.microsoft.com/en-us/azure/app-service/overview-manage-costs>

ii. Plan and Manage your Azure Costs

<https://docs.microsoft.com/en-us/learn/modules/plan-manage-azure-costs/>

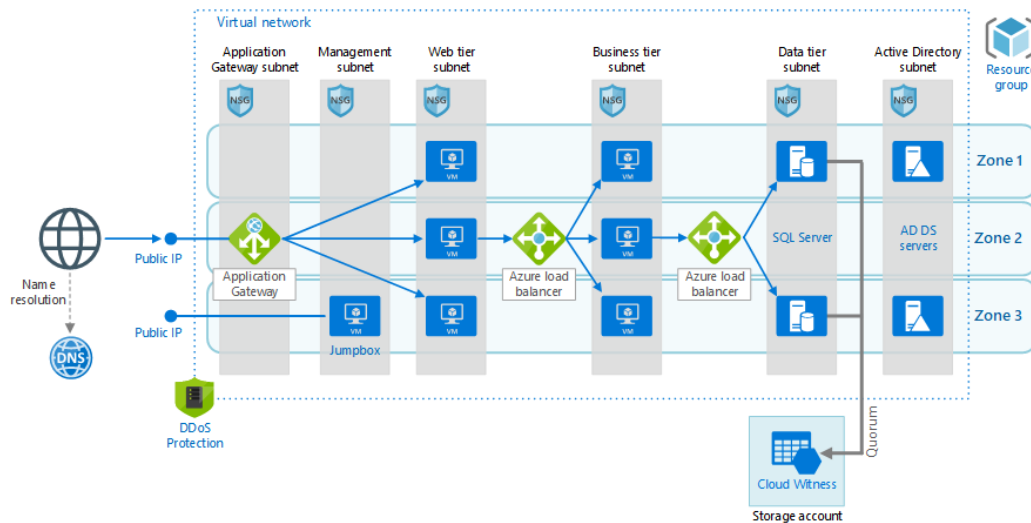
ii. Configure app service plans

<https://docs.microsoft.com/en-gb/learn/modules/configure-app-service-plans/>

Exercise:

In this task, you have to calculate the cost for the following windows N-tier applications on Azure. At this stage, you don't need to understand each terminology used in the following system.

As we can see, there are **five computation resources** needed to build this system: web tier, Business tier, Data tier, Jump box and AD server.



Q1) Calculate Web Tier VM machine cost for all VM.

To calculate the cost for Web Tier, we have to use the Microsoft Azure price calculator.

<https://azure.microsoft.com/en-us/pricing/calculator/>

Pricing calculator

Calculate your estimated hourly or monthly costs for using Azure.

[Get started with Azure](#)



Products Example scenarios Saved estimates FAQs

Anum Abeera
Abeera-A@ulster.ac.uk
Ulster University

Select a product to include it in your estimate.

Search products

- Popular
- Compute
- Networking
- Storage
- Web
- Mobile
- Containers
- Databases
- Analytics
- AI + machine learning
- Internet of Things
- Integration
- Identity

Virtual Machines

Provision Windows and Linux VMs in seconds

[Add to estimate](#)

Storage Accounts

Durable, highly available, and massively scalable cloud storage

[Add to estimate](#)

Azure SQL Database

Build apps that scale with managed and intelligent SQL database in the cloud

[Add to estimate](#)

App Service

Quickly create powerful cloud apps for web and mobile

[Add to estimate](#)

Azure Cosmos DB

Build or modernize scalable, high-performance apps

[Add to estimate](#)

Azure Kubernetes Service (AKS)

Deploy and scale containers on managed Kubernetes

[Add to estimate](#)

Azure Functions

Execute event-driven serverless code functions with an end-to-end development experience

[Add to estimate](#)

Azure AI services

Add cognitive capabilities to apps with APIs and AI services

[Add to estimate](#)

Microsoft Cost Management

Monitor, allocate, and optimize cloud costs with transparency, accuracy, and efficiency

[Add to estimate](#)

As you need to calculate Virtual Machine, please click on the Virtual machines tab.

Virtual Machines 1 D2 v3 (2 vCPUs, 8 GB RAM) x 730 Hours (Pay as yo... Upfront: \$0.00 Monthly: \$152.62

Virtual Machines

REGION: West US OPERATING SYSTEM: Windows TYPE: (OS Only) TIER: Standard

CATEGORY: All VM SERIES: All INSTANCE: D2 v3: 2 vCPUs, 8 GB RAM, 50 GB Temporary storage, \$0.209/hour

VIRTUAL MACHINES 1 x 730 Hours

Savings Options

Save up to 72% on pay-as-you-go prices with 1-year or 3-year Reserved Virtual Machine Instances. Reserved Instances are great for applications with steady-state usage and applications that require reserved capacity. [Learn more about Reserved VM Instances pricing.](#)

Compute (D2 v3) ☒ Pay as you go ☐ 1 year reserved (~32% discount) ☐ 3 year reserved (~57% discount) \$85.41 Average per month (\$0.00 charged upfront)	OS (Windows) ☒ License included ☐ Azure Hybrid Benefit \$67.16 Average per month (\$0.00 charged upfront)	= \$152.57 Average per month (\$0.00 charged upfront)
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Managed Disks \$0.00

Storage transactions \$0.05

Bandwidth \$0.00

The default region is East US (or West US), and the default currency is US dollars. Please choose your desired currency and region. Change the OS and other configurations to the requirements needed in the table. (hint: Change OS to Linux)

Lab Task:

Complete the cost for the given Business Tier table.

Resource	Region	# Hours	CPU	Memory	Temp Storage	Managed Disk	Os Type	Storage Transaction	Bandwidth Outbound	Upfront cost (\$)	Monthly Cost (\$)
VM1	UK West	500	8 Cores	56 GB RAM	605GB	Standard – HDD S4	Ubuntu	0	20GB		
VM2	UK South	744	1 Core	.75 GB RAM	20	Standard – SSD 8 GiB x 2	SQL Server	0	15GB		

Resource	Region	Type	Service	Compute	# Hours	Storage	Backup Storage	Long Term Retention	Support	Upfront cost (\$)	Monthly Cost (\$)
Azure SQL DB x 3	UK West	Single DB	Business Critical	8 vCore	700	16 GB	None	None	Standard - MCA		

* Export the quote in Excel



Throughout this module, attempt to exclusively use free tier services or serverless/consumption provisions.

To this end, regularly check the consumption of your credit and ensure you shut down services and resources when they are not being used. Failure to do so may incur a large bill that you may be liable for.

Expenditure may be monitored at :

- <https://www.microsoftazuresponsorships.com/Balance>
- https://portal.azure.com/#blade/Microsoft_Azure_GTM/ModernBillingMenuBlade/BillingAccounts

Note: A high quality coursework submission may be produced by exclusively using free tier services which will incur no costs/wont consume credit.